Support VM/WSL Setup

Laboratórios de Sistemas e Serviços

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Introduction: Choose Your Environment

To ensure a consistent workspace, you must set up a Linux environment. Choose **one** of the following options based on your operating system:

1. **VirtualBox:** Best for Intel/AMD Windows & Linux PCs. (Standard Choice).
2. **VMware Workstation:** High-performance alternative for Windows/Linux.
3. **UTM:** The **required** choice for Apple Silicon Macs (M1/M2/M3 chips).
4. **WSL (Windows Subsystem for Linux):** Best for advanced Windows users who prefer terminal integration over a full GUI VM.

Part 1: Download the Class Disk 📀

For **VirtualBox, VMware, and UTM**, you need the pre-configured Debian disk.

1. **Download:** Download Debian VM Disk (.vdi)
2. **Save:** Save to `Downloads` or `Documents`.

> ⚠️ **format Warning:** This file is in `.vdi` format (VirtualBox native). * **VirtualBox:** Use as is. * **VMware/UTM:** You may need to convert it or install a fresh Debian ISO if the import fails (See Appendix A).

Option A: VirtualBox (Standard for Windows/Intel Mac) 💻

Do not use this on Apple Silicon (M1/M2/M3) Macs. Use Option C (UTM) instead.

1. Installation

1. **Download:** Go to virtualbox.org/wiki/Downloads.
2. **Install App:** Download and run the installer for your OS ("Windows hosts" or "macOS / Intel hosts").
3. **Install Extension Pack:** Download the "Oracle VM VirtualBox Extension Pack" from the same page and double-click it to install.

2. Create VM

1. Open VirtualBox, click **New**.
   - **Name:** `Debian IEI`
   - **Type:** `Linux` / **Version:** `Debian (64-bit)`
2. **RAM:** 4096 MB (4 GB) recommended.
3. **Hard Disk:** Select **"Use an Existing Virtual Hard Disk File"**.
4. Click the folder icon, navigate to the `.vdi` file you downloaded, and select it.
5. Click **Create**.

3. Drivers (Guest Additions)

1. Start the VM (Login: `student` / `password`).
2. In the VM menu: **Devices > Insert Guest Additions CD image**.
3. Open Terminal in VM and run:

```
sudo apt update
sudo apt install build-essential dkms linux-headers-$(uname -r)
sudo mkdir -p /mnt/cdrom
sudo mount /dev/cdrom /mnt/cdrom
sudo /mnt/cdrom/VBoxLinuxAdditions.run
sudo reboot
```

Option B: VMware Workstation (Windows/Linux) 🚀

VMware Workstation Pro is now **free for personal use** and often faster than VirtualBox.

1. Installation

1. **Download:** Create a Broadcom account and download **VMware Workstation Pro** (Windows) or **VMware Fusion** (Mac - Intel only).
2. **Install:** Run the installer. Select "I want to license for Personal Use" (no key required).

2. Create VM (Importing VDI)

Note: VMware uses `.vmdk`*. You can try selecting the* `.vdi` *by choosing "All Files", but conversion is safer (See Appendix A).*

1. Click **"Create a New Virtual Machine"** > **Custom (Advanced)**.
2. **Guest OS:** Linux > Debian 12 (64-bit).
3. **Disk:** Choose **"Use an existing virtual disk"**.
4. Browse for your converted `.vmdk` file (or try the `.vdi`).
5. When asked to **"Convert"** the format, click **Yes**.

3. Drivers (Open-VM-Tools)

VMware uses open-source drivers. 1. Start VM (Login: `student / password`). 2. Open Terminal and run: `bash sudo apt update sudo apt install open-vm-tools-desktop sudo reboot` *This enables copy/paste, drag-and-drop, and auto-resizing.*

Option C: UTM (Apple Silicon Mac M1/M2/M3) 🍎

This is the only performant option for modern Macs.

1. Installation

1. Download **UTM** from mac.getutm.app (Free) or the Mac App Store (Paid/Donation).
2. **Important:** The class disk is `.vdi` (x86 architecture). Emulating x86 on Apple Silicon is **slow**.
   - *Recommendation:* It is highly recommended to **install a fresh ARM64 Debian** instead of using the class disk.

- *If you must use the class disk:* You need to convert it to `.qcow2` first (See Appendix A) and accept slow emulation speeds.

2. Create VM (Fresh Install Method - Recommended)

1. Download the **Debian ARM64 ISO** from debian.org.
2. Open UTM > **Create a New Virtual Machine**.
3. **Virtualize** (Fast) > **Linux**.
4. **Boot Image:** Select the Debian ARM64 ISO.
5. Follow the installer steps.

3. Drivers (SPICE Agent)

To get clipboard sharing and dynamic resolution: 1. Open Terminal in VM. 2. Run: `bash sudo apt update sudo apt install spice-vdagent sudo reboot`

Option D: Windows Subsystem for Linux (WSL) 🪟

Runs Linux natively alongside Windows. High performance, but no "Virtual Desktop" window by default.

1. Installation

1. Open **PowerShell** as Administrator.
2. Run:

   ```
   wsl --install
   ```

3. **Reboot** your computer.
4. After reboot, "Ubuntu" will open. Create a username/password.

2. Optimize Networking (Mirrored Mode)

To make WSL behave like a real PC on your network (fixing many connectivity issues):

1. In Windows, press `Win+R`, type `%UserProfile%`, and hit Enter.
2. Create a file named `.wslconfig` (make sure it's not `.txt`).
3. Paste this content:

```
[wsl2]
networkingMode=mirrored
dnsTunneling=true
autoProxy=true
```

4. Restart WSL: `wsl --shutdown` in PowerShell.

3. GUI Apps

WSL supports GUI apps out of the box. Try running:

```
sudo apt update && sudo apt install gedit nautilus
gedit
```

(The editor should appear in a window).

Appendix A: Converting the Disk (.vdi) 🔧

If you are not using VirtualBox, you may need to convert the downloaded `.vdi` file.

For VMware (VDI -> VMDK):

Requires VirtualBox to be installed. Run in Command Prompt:

```
"C:\Program Files\Oracle\VirtualBox\VBoxManage.exe" clonemedium disk "C:\Path\T
```

For UTM (VDI -> QCOW2):

Requires qemu (Install via Homebrew: `brew install qemu`).

```
qemu-img convert -f vdi -O qcow2 Input.vdi Output.qcow2
```